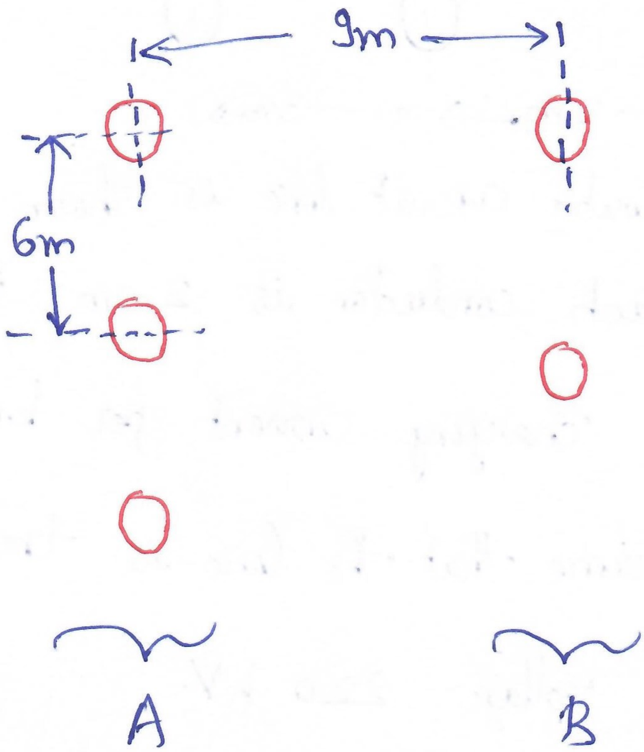
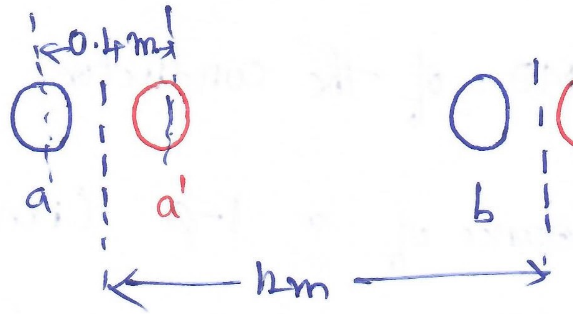
Assignment - 1

1. A conductor consist of seven identical strands each having a radius of 'r'. Determine the factor by which 'r' should be multiplied to find the self GMD of the conductor.

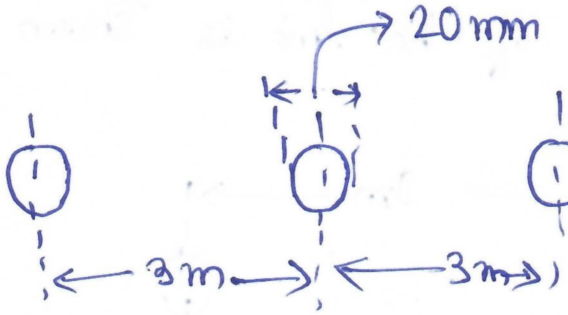
2. Determine the inductance of a 1-ϕ transmission line consisting of three conductors of 2.5mm radii in the 'go' conductor and two conductors of 5mm radii in the return conductor. The configuration of the line is shown below.



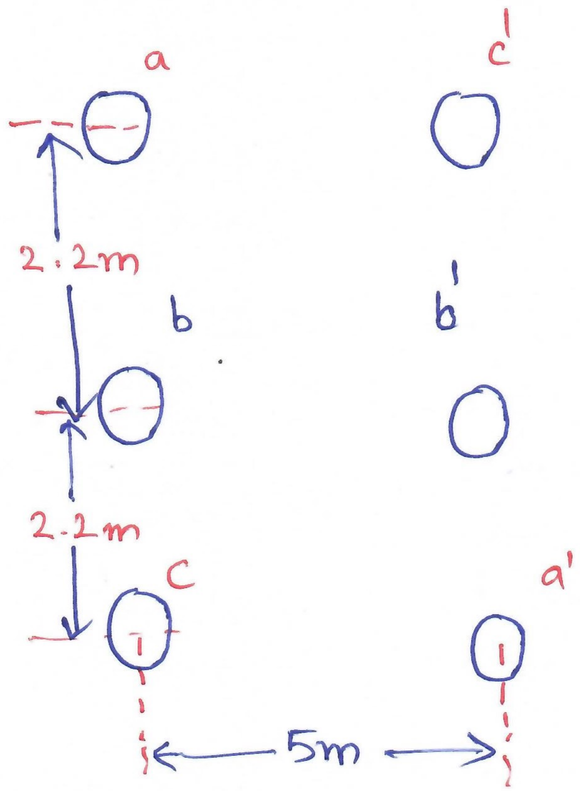
(3) Determine the inductance per km of a 3-ϕ transmission line having conductors per phase and arranged as shown below.



(4) Determine the capacitance and charging current per unit length of the line when the arrangement of the conductor is as shown below. The operating voltage is 132 kV.



(5) A three phase double circuit line is shown below. The diameter of each conductor is 2 cm. Determine the capacitance and charging current per km length of the line, Assume that the line is transposed and the operating voltage 220 kV.



(6) Why ACSR conductors are preferred for transmission and distribution lines.?